

Rayane El Mselmi

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Second-year Computer Science student at Durham University (Expected 2027) with hands-on experience building data-driven full-stack applications, including REST APIs, SQL, and backend systems using Node.js.

EDUCATION

Durham University — BSc Computer Science (Expected 2027)

- Year 1: 2:1

Durham University International Study Centre: Computer Science Foundation Year (*82% overall*)

EXPERIENCE

Software Engineering Intern (Full-Stack) | COPAG, Morocco

Mar 2026 – Apr 2026

- Built a full-stack data platform using Node.js, Express and SQLite to process and analyse logistics operations data
- Designed and implemented a REST API with dynamic filtering (driver, route, vehicle ID) to query operational data
- Developed SQL-based KPI calculations (average waiting time, unloading duration, volume per driver) using grouping logic to avoid double counting across multi-row operations
- Built an interactive dashboard using React and Recharts to present operational insights through charts and tabular views
- Implemented basic API testing with Jest and Supertest to validate endpoints and ensure reliable responses

Lead Web Developer | Critique Durham, Durham University

Mar 2026 – Present

- Leading development of a production-ready full-stack journal platform with submission and editorial workflows
- Designed and implemented a submission and approval workflow, handling article uploads and editorial review
- Built backend API routes and database models to manage submissions, metadata and publication status
- Developed dynamic pages to display approved articles (PDF-based) and ensure controlled content visibility
- Collaborated with editorial stakeholders to translate requirements into functional system features and workflows

PROJECTS

StudySync | Full-Stack Task Manager (React, Node.js, PostgreSQL) [Link](#)

- Built and deployed a full-stack task management application with JWT authentication and protected API routes
- Implemented task CRUD operations and dashboard summaries using Express and PostgreSQL

EventHub Pro | Events REST API (Express, Prisma/SQLite) [Link](#)

- Developed a paginated REST API with search and filtering
- Implemented security middleware for API protection

Maze Generation Algorithms | Benchmarking Study (Python, NumPy, Matplotlib)

- Benchmarked DFS, Prim's and Kruskal algorithms across grid sizes
- Analysed results using CSV datasets and Matplotlib visualizations

SKILLS

Languages: Python, Java, JavaScript, HTML/CSS

Backend & APIs: Node.js, Express, REST APIs

Databases: PostgreSQL, SQLite, Prisma

Testing & Tools: Jest, Supertest, Git, Bootstrap, NumPy, Pandas, Matplotlib

CERTIFICATIONS

- **Data Fundamentals** | IBM SkillsBuild (2026)
- **Cloud Computing Fundamentals** | IBM SkillsBuild (2026)

TECHNICAL ACTIVITIES

Technical Problem-Solving (CTF Assessment) | 80+/100 through VM-based debugging and system analysis challenges

LANGUAGES

Arabic (Fluent) | English (C1) | French (C1) | Spanish (A1–A2)